

REVIEW

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Nutrition and longevity – diet in centenarians

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Abstract

Background Nutrition plays a central role in the biological mechanisms that shape aging, health span, and longevity. Micronutrients—including vitamins, trace elements, and polyphenols—support genomic stability, mitochondrial integrity, and antioxidant defense, while dietary patterns rich in plant-based foods modulate inflammation, metabolic regulation, and epigenetic processes. Centenarian populations consuming Mediterranean, Okinawan, Nordic, and Nicoyan diets offer a natural model for understanding how nutrient-rich, minimally processed foods, moderate caloric intake, and balanced lifestyles interact with molecular pathways to extend functional life.

Main Body This review synthesizes current evidence on how micronutrients influence DNA repair, oxidative stress reduction, and mitochondrial protection, particularly through the actions of vitamins C and E, niacin-dependent PARP activity, folate-mediated methylation, and metal cofactors involved in antioxidant enzymes. Plant-based diets rich in fiber and polyphenols enhance microbial diversity and promote beneficial taxa such as *Akkermansia* and *Bifidobacterium*, supporting gut barrier integrity and immune balance. Caloric restriction and intermittent fasting activate nutrient-sensing pathways, including AMPK and sirtuins, reduce mTOR activity, and stimulate autophagy, collectively improving cellular resilience. Findings from centenarian regions highlight the convergence of lifestyle, nutrition, and cultural practices that reduce systemic inflammation, maintain metabolic flexibility, and support healthy aging trajectories.

Conclusions Diet emerges as a decisive modifiable determinant of lifespan and health span. The convergence of molecular nutrition, microbiome composition, and traditional dietary habits underlies the exceptional longevity observed in centenarian populations. Future research should integrate nutrigenomics, metabolomics, and microbiome profiling to clarify causal mechanisms and guide precision nutrition strategies for aging societies.

Keywords Centenarians, nutrition, Dietary patterns, Gut microbiota, Healthy aging

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Introduction

Aging is a complex process shaped by the interaction between genetics, environmental, and lifestyle factors [1]. While heredity sets a baseline for longevity, nutrition emerges as one of the most influential, modifiable determinants of healthy aging. Beyond sustaining basic physiological needs, dietary patterns actively regulate molecular pathways associated with healthspan, including oxidative stress, inflammation, cellular repair mechanisms, and mitochondrial function [2, 3].

Centenarians offer a unique model for understanding how nutrition supports long life. Their dietary habits—frequently aligned with the Mediterranean diet (MD) and observed in so-called Blue Zones—are characterized by a high intake of plant-based foods and unsaturated fats, patterns consistently associated with increased lifespan and reduced risk of chronic disease [4]. These populations illustrate that nutrition may not only prolong survival but also preserve functional capacity and quality of life in advanced age.

At the biological level, specific nutrients modulate core mechanisms of aging: antioxidants mitigate oxidative damage; omega-3 fatty acids and polyphenols attenuate chronic inflammation; and bioactive compounds such as zinc, folate, and spermidine support DNA repair, autophagy, and mitochondrial homeostasis [5]. In parallel, dietary strategies including intermittent fasting (IF) and ketogenic approaches influence metabolic flexibility and cellular recycling pathways, further underscoring the close relationship between diet and longevity [6, 7].

This review synthesizes current evidence on the role of nutrition in aging, examining the gut microbiota, energy intake, macro- and micronutrients, dietary patterns, nutraceuticals, and evidence-based nutritional strategies. By framing centenarian diets as real-world models,

this work aims to identify actionable nutritional principles that may contribute to extending both lifespan and healthspan.

Nutrition and longevity

Micronutrients, such as vitamins and trace elements, are vital for immune function, metabolism, and tissue repair [8] (Fig. 1). Nutrients like zinc, magnesium, selenium, and vitamins C and E are essential for maintaining genomic integrity by supporting DNA repair and mitigating oxidative stress [12]. Water-soluble vitamins primarily function as coenzymes or precursors to enzymes, assisting in various biochemical reactions within the body [13, 14], as detailed in Table 1.

Given that plant-based diets provide essential micronutrients, they have the potential to influence key biological processes involved in aging and longevity (Fig. 1). A clear example is the MD, known for its antioxidant and anti-inflammatory properties, which has been linked to longer telomeres, while high-fat diets negatively affect telomere stability [40]. Epigenetic alterations, including changes in DNA methylation and histone modifications, further contribute to aging [10, 41]. Diets high in saturated fats induce methylation changes, accelerating aging, while certain vitamins modulate gene transcription and maintain cellular health [42, 43]. Autophagy, the cellular process of degrading and recycling damaged components, is also crucial for aging. Nutrients like spermidine, found in whole grains and vegetables, stimulate autophagy and extend lifespan [44].

Mitochondrial function, vital for energy production, deteriorates with age due to DNA mutations and oxidative stress. Omega-3 fatty acids, B vitamins, and other micronutrients support mitochondrial integrity, reducing chronic inflammation and tissue damage [41, 45–47].

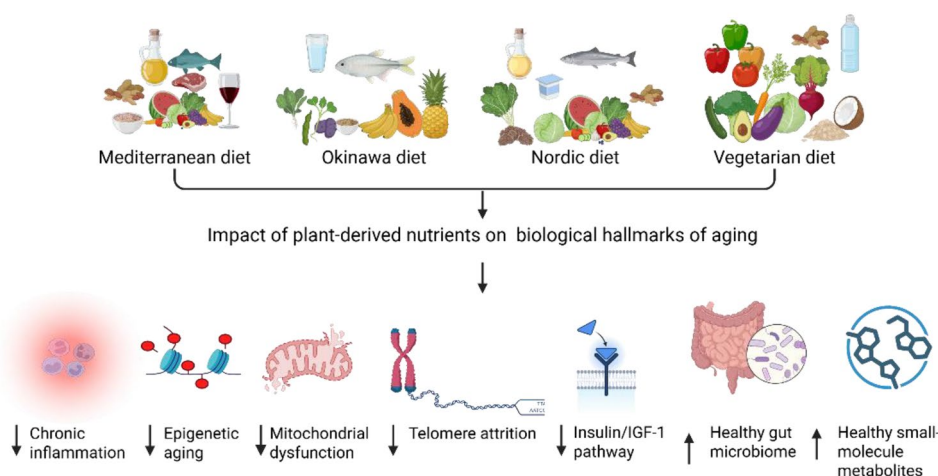


Fig. 1 Nutritional approaches for supporting healthy aging and longevity, highlighting the Mediterranean, Okinawa, Nordic, and Vegetarian diets. These dietary patterns influence key hallmarks of aging, including inflammation, epigenetic aging and longevity-related markers [9–11]. Abbreviations: IGF-1: insulin-like growth factor 1

Table 1 Role of micronutrients in biochemical reactions

Micronutrient(s)	Function in Genomic Stability	Effects of Deficiency
Folate and Vitamins B ₂ , B ₆ , B ₁₂	DNA methylation, synthesis of dTTP from dUMP, effective folate recycling [15–18].	Leads to uracil being incorporated into DNA, increased chromosome instability, and reduced DNA methylation [17–21].
Magnesium	Cofactor for several DNA polymerases in various repair mechanisms, supports microtubules, chromosome alignment [22, 23].	Poor DNA replication accuracy, impaired repair mechanisms, and chromosomal segregation defects [24].
Zinc	Cofactor for enzymes like Cu/Zn superoxide dismutase and others involved in DNA repair and genome protection [25].	Elevated oxidative damage to DNA, chromosome instability, increased DNA breaks [26].
Vitamin C, Vitamin E	Protects DNA and lipids from oxidative stress [27, 28].	Higher rates of DNA strand breaks, chromosome instability, increased DNA oxidation [27, 28].
Iron	Integral to ribonucleotide reductase and mitochondrial electron transport proteins [29, 30].	Compromised DNA repair, heightened vulnerability of mitochondrial DNA to oxidative damage [9, 31].
Niacin	Required by PARP enzymes essential for DNA repair and maintaining telomere integrity [32–34].	Buildup of unrepaired DNA nicks, chromosomal rearrangements, higher sensitivity to mutagens [35, 36].
Manganese	Cofactor in mitochondrial Mn superoxide dismutase [37].	Mitochondrial DNA damage, weaker defense against nuclear DNA damage from radiation [38, 39].

Abbreviations: dTTP: deoxythymidine triphosphate; dUMP: deoxyuridine monophosphate; DNA: deoxyribonucleic acid; Cu: copper; Zn: zinc; PARP: Poly (ADP-ribose) polymerase; Mn: manganese

Nutrient-sensing pathways, such as mTOR and AMPK, regulate growth and metabolism and can be modulated by dietary interventions to promote longevity. Inflammaging, a persistent low-grade inflammation linked to age-related diseases, can be mitigated by caloric restriction (CR) and senolytic treatments like fisetin [46, 47].

Gut microbiota and longevity

The gut microbiome plays a crucial role in maintaining health by regulating digestion, immune defense, and synthesizing essential compounds like short-chain fatty acids (SCFAs), amino acids, and vitamins [8, 48]. Disruptions in microbiome communication, known as dysbiosis, contribute to chronic conditions such as obesity, type 2 diabetes, cardiovascular diseases, and neurodegeneration,

which negatively affect healthspan and longevity [49–52]. Dysbiosis affects gut permeability, overstimulating immune responses and triggering chronic inflammation, which damages epithelial cells and disrupts overall health [52]. Certain bacteria, such as segmented filamentous bacteria, can help reinforce gut barrier integrity [53, 54]. SCFAs, particularly butyrate, typically present in the colonic lumen at concentrations ranging from approximately 10 to 20 mM, produced by beneficial microbes such as *Faecalibacterium prausnitzii* and *Akkermansia muciniphila*, play a central role in maintaining intestinal epithelial integrity, serving as an energy source for colonocytes and modulating local and systemic immune responses [55]. These microbes’ levels decline with age, particularly in those with frailty or chronic inflammation. Age-related inflammation, or “inflammaging,” is characterized by elevated markers like interleukins (IL-1, IL-6) and tumor necrosis factor (TNF-α), which plays a role in the development of disorders such as Alzheimer’s disease and atherosclerosis [56].

Gut microbiome composition changes with age, with microbial diversity often declining in older adults. However, studies of centenarians show a reduction in dominant bacteria like *Bacteroides*, alongside an increase in beneficial bacteria such as *Bifidobacterium* and *Akkermansia*. These microbes are associated with health benefits, including improved metabolic and cognitive function. For instance, *Akkermansia muciniphila* supports intestinal integrity and mitigates the effects of high-fat diets, while members of the *Christensenellaceae* family are associated with improved metabolic profiles through their involvement in SCFAs production, modulation of host lipid metabolism, and attenuation of low-grade systemic inflammation, and are consistently linked to lower heart disease risk and reduced type 2 diabetes [56, 57].

Microbial metabolites, such as indole from tryptophan metabolism, promote gut health and reduce inflammation, which are linked to longevity. Diets rich in fiber, polyphenols, and fermented foods can support microbial diversity and help prevent dysbiosis, while high-fat, high-sugar diets contribute to chronic inflammation. CR, IF, and prebiotics have also shown potential in reshaping the microbiome and reducing inflammation [58–60]. Targeting the microbiome with personalized interventions could delay age-related diseases [61], offering promising strategies for extending healthspan and promoting longevity [62–65].

Total energy intake, macronutrients and micronutrients and longevity

Calorie restriction and energy balance

CR, the reduction of nutrient intake without malnutrition, has been shown to extend lifespan and slow aging

[66]. It reshapes the gut microbiome—enhancing beneficial bacteria like *Lactobacillus*—and activates metabolic pathways such as AMPK, SIRT1, and SIRT3, which inhibit mTOR signaling, reduce oxidative stress, and lower inflammation [67].

CR also improves insulin sensitivity and glucose regulation [68, 69], potentially limiting protein glycation—a mechanism linked to healthier aging and observed in centenarians [70]. Additionally, CR reduces the risk of age-related diseases like hypertension, type 2 diabetes,

and neurodegenerative conditions [71]. Maintaining energy balance—adequate but not excessive caloric intake—is essential, as both overnutrition and undernutrition impair immunity and reduce longevity.

Carbohydrates

Carbohydrates influence aging depending on both their type and quantity. The “carbohydrate quality” concept distinguishes between processed, high-glycemic carbs, and complex, fiber-rich carbs (Table 2) [67]. A study by Yi et al. found that high-quality carbohydrates, particularly fiber-rich foods, slow biological aging, as indicated by epigenetic markers [75]. Another difference to note is the glycemic index (GI), which impacts metabolic health; high-GI foods trigger rapid glucose spikes, promoting insulin resistance, inflammation, and type 2 diabetes, while low-GI foods, especially fiber-rich fruits, help stabilize blood sugar, reduce disease risk, and support healthy aging [67] (Table 2).

Protein

Protein is vital for tissue repair, enzyme function, immune health, and preserving muscle mass with age. However, high intake—especially from animal sources—can activate the mTOR pathway, accelerating aging and age-related diseases. Protein and amino acid restriction have been linked to increased longevity by lowering IGF-1 levels, reducing cancer risk, and activating FOXO3A, which supports anti-inflammatory responses and genomic stability [76, 77].

Moderate intake from plant sources may be more beneficial, as plant proteins are lower in leucine (a key mTOR activator) and richer in fiber, vitamins, and minerals. Diets emphasizing plant proteins can reduce chronic disease risk and promote longevity. Additionally, evenly spaced protein consumption throughout the day helps maintain muscle mass and prevent sarcopenia in older adults [67].

Fat

The type of fat consumed is more important than the total fat intake in influencing aging and chronic disease risk [67]. Unsaturated fats—particularly from plant sources and marine oils—are linked to reduced mortality, improved cardiovascular and cognitive health, and lower risk of neurodegenerative diseases like Alzheimer’s [67, 78]. Polyunsaturated fats, especially omega-3s, and monounsaturated fats (MUFAs) from olive oil, avocados, and nuts help lower LDL cholesterol, reduce inflammation, and support vascular function [79]. Studies such as the HAPIEE cohort have associated higher omega-3 and MUFA intake with better aging outcomes [80]. Conversely, saturated and trans fats contribute to brain aging, insulin resistance, and cognitive decline, with trans fats

Table 2 Comparison of processed and complex carbohydrates regarding metabolic health and aging

Type of Carbohydrate	Definition/Criteria	Sources	Effects	Impact on Aging
Processed Carbohydrates	Highly refined carbohydrates, low fiber content, high glycemic index	Refined sugars [72]	Contribute to obesity, insulin resistance, and oxidative stress [72]	Accelerate aging [72]
Complex Carbohydrates	Minimally processed carbohydrates, high fiber content, lower glycemic index	Legumes, vegetables, whole grains (Naturally rich in fiber) [73, 74]	Enhance digestion, reduce inflammation, improve insulin sensitivity, Fiber supports gut microbiome and lowers risk of chronic diseases (e.g., cancer, cardiovascular disease) [73]	Support longevity and lifespan [74]

promoting systemic inflammation and oxidative stress that accelerate aging and increase cardiovascular risk [81].

Micronutrients and other substances

Micronutrients play a vital role in healthy aging. B vitamins and omega-3 fatty acids support cognitive and psychological health; vitamins B6, B12, and folate help lower homocysteine levels, which are linked to cognitive decline. Vitamin D and omega-3s reduce cardiovascular risk, while vitamin E may protect against oxidative damage [67].

Plant-derived compounds like polyphenols, flavonoids, and carotenoids—found in fruits, vegetables, and tea—promote longevity by reducing inflammation and oxidative stress. Diets rich in these phytochemicals, particularly plant-based diets, are associated with lower rates of chronic diseases and increased lifespan [67].

Dietary patterns and longevity

The Mediterranean diet

The MD, characterized by abundant fruits, vegetables, whole grains, legumes, fish, and olive oil, remains the most extensively studied longevity model [82]. A recent meta-analysis reported that higher adherence to the MD was associated with a 23% reduction in all-cause mortality and a 27% decrease in cardiovascular-related deaths [83, 84]. This effect is largely attributed to heart-healthy components such as olive oil, fish, and whole grains [85]. Olive oil improves cholesterol and reduces inflammation, while fish-derived omega-3 fatty acids lower blood pressure (BP) and prevent heart disease, key contributors to cardiovascular longevity [86]. The MD's anti-inflammatory and antioxidant properties counters chronic inflammation (“inflammaging”), a major contributor to many age-related diseases, including Alzheimer's, type 2 diabetes, and atherosclerosis. Dietary antioxidants such as lycopene, polyphenols, and flavonoids neutralize reactive oxygen species and modulate inflammatory cytokines [3, 5].

Additionally, the MD supports gut health through high fiber intake from whole grains, fruits, and vegetables nourishes beneficial bacteria, enhancing digestion, immunity, and metabolism. Olive oil and nuts enhance microbial balance, while fiber-rich fruits help regulate blood sugar and reduce insulin resistance. Cognitive protection has been linked to MD adherence [87, 88]. In the NU-AGE Project, where higher intake of omega-3s and polyphenols correlated with slower epigenetic aging and improved memory [89, 90].

Sardinia's Blue Zone exemplifies a Mediterranean-derived pattern emphasizing legumes (chickpeas, fava beans), whole grains, moderate dairy, and wild greens [91, 92]. Moderate consumption of polyphenol-rich

beverages, *Cannonau* wine and coffee, provides additional vascular and cognitive protection [93–99]. Collectively, these synergistic mechanisms position the MD as both a health-promoting and sustainable dietary model [100].

Vegetarian and other plant-based diets

Plant-centered diets share with the MD a focus on minimally processed fiber- and polyphenol-rich foods. The Nordic diet emphasizes locally sourced, plant-based foods such as whole grains, root vegetables, berries, and legumes, along with fatty fish like salmon, while limiting processed foods and red meat [101]. Evidence shows improvements in cardiometabolic markers, including LDL cholesterol, apolipoprotein B, insulin levels, and systolic BP. Its high fiber content enhances insulin sensitivity and gut microbiome health, while antioxidant-rich berries help combat oxidative stress and promote long-term well-being [102, 103].

The Okinawan traditional diet derives about 80% of energy from plant sources, mainly sweet potatoes, soy, and vegetables, with minimal animal fat and sodium [104]. The cultural practice *hara hachi bu* (eating until 80% full) promotes mild CR, which has been linked to reduced oxidative damage and extended lifespan [67]. Combined with a low omega-6/omega-3 ratio and high antioxidant intake, this pattern correlates with a lower chronic disease prevalence [104].

The Nicoyan diet (Costa Rica) centers on minimally processed staples like maize, beans, and squash—the “Mesoamerican triad”—along with potatoes, fruits, and legumes, offering high fiber and antioxidants with low energy density [105]. Similarly, Seventh-day Adventists in Loma Linda, California, consume diets rich in legumes, whole grains, and vegetables, often vegetarian or vegan, while avoiding alcohol, caffeine, and unclean meats [106–109]. These practices are associated with [107] reduced IGF-1 levels, diminished oxidative stress, and greater longevity [110].

Vegetarian and vegan diets, which further restrict or eliminate animal products [111], are linked to lower BP, LDL cholesterol, and body weight [112, 113], as well as enhanced insulin sensitivity, improved lipid profiles, and greater microbial diversity [114, 115]. The exclusion of animal proteins such as methionine and leucine may promote longevity by reducing cellular growth, inflammation, and oxidative damage [116]. Although stricter adherence may require nutrient supplementation of vitamin B12, iron, calcium, and protein, large-scale epidemiological studies consistently support plant-based diets for improving metabolic health, reducing inflammation, and extending lifespan [110, 117, 118].

Overall, plant-based patterns, like the Nordic, Okinawan, Adventist, Nicoyan, vegetarian, and vegan diets,

emphasize whole, nutrient-dense foods that support a healthy gut microbiome, lower inflammation, and improve metabolic function [67]. Collectively, they promote healthy aging and longevity.

Intermittent fasting and longevity

IF protocols (alternate-day or time-restricted feeding), modulate key aging pathways (Fig. 2) [119]. Clinical and experimental consistently reveal that IF improves lipid profiles, lowers fasting insulin, and enhances insulin sensitivity, and reduces body weight and blood pressure, helping mitigating age-related cardiovascular decline [120, 121]. Hoddy et al. [122], reported additional reductions in total cholesterol (13%), LDL cholesterol (15%), triglycerides (22%), and oxidative stress markers such as 8-isoprostane.

Population studies mirror these findings. In Abruzzo, Italy, elderly individuals practicing prolonged nightly fasting (“sdijuno”) show favorable metabolic and immune profiles linked to exceptional longevity [123]. In Ikaria, Greece, religious fasting combined with a MD are associated with better cognitive health and a lower prevalence of dementia [124, 125].

Mechanistically, IF reduces growth hormone and IGF-1 signaling, activates AMPK, and inhibits mTOR, thereby enhancing autophagy via TFEB, TFE3, and S6K. Concurrent activation of sirtuins (SIRT1, SIRT3) promotes

PGC-1 α and FOXO pathways, supporting mitochondrial biogenesis and cellular stress resistance. IF also reduces inflammation and oxidative stress by modulating NF- κ B and Nrf2 activity [3, 119, 126, 127].

In summary, IF appears to be a multifaceted approach that supports metabolic balance, circadian regulation, immune function, and healthy gut microbiota, ultimately promoting healthy aging and reducing disease risk [128].

Dietary patterns fostering longevity converge on shared principles—plant predominance, healthy fats, minimal sodium, moderated caloric intake, and synchronized meal timing. Together, they modulate inflammatory, metabolic, and epigenetic processes that underpin healthy aging (Table 3).

Evidence-based nutritional strategies for achieving longevity and reaching 100 years of age

Dietary patterns associated with exceptional longevity and centenarian populations

Across global longevity hotspots, dietary patterns share common features linked to exceptional life expectancy. These diets are predominantly plant-based, emphasizing whole grains, fruits, vegetables, and legumes, with modest fish and poultry intake and minimal red meat, saturated fat, and ultra-processed products [131, 132]. Observed sodium intake in long-lived populations averages approximately ~1.6g/day, which is below World

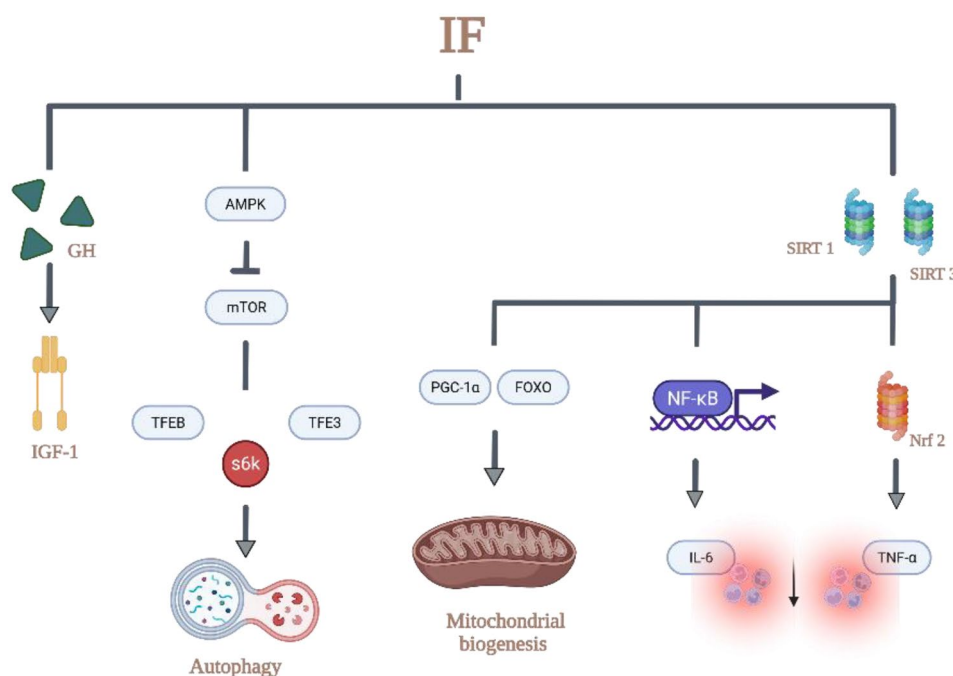


Fig. 2 Nutrient-sensing and stress-responsive signaling pathways influenced by intermittent fasting. IF: intermittent fasting; GH: growth hormone; IGF-1: insulin-like growth factor 1; AMPK: AMP-activated protein kinase; mTOR: mammalian target of rapamycin; TFEB: transcription factor EB; TFE3: transcription factor E3; S6K: ribosomal protein S6 kinase; SIRT1/3: sirtuin 1/sirtuin 3; PGC-1 α : peroxisome proliferator-activated receptor gamma coactivator 1-alpha; FOXO: forkhead box O; NF- κ B: nuclear factor kappa-light-chain-enhancer of activated B cells; IL-6: interleukin 6; Nrf2: nuclear factor erythroid 2-related factor 2; TNF- α : tumor necrosis factor alpha

Table 3 Common features of longevity-associated diets

Feature	Typical Components	Physiological Effects
Predominantly plant-based [67, 83–85, 105–109]	Whole grains, legumes, fruits, vegetables, nuts	↓ inflammation, improved insulin sensitivity
Healthy fats [85, 102,103]	Olive or rapeseed oil, nuts, fish	↓ LDL-C, improved vascular function
Moderate caloric intake [104, 127]	Portion control (e.g., <i>hara hachi bu</i>)	↓ oxidative stress, improved metabolic efficiency
Low sodium intake [129, 130]	< 2 g/day, use of potassium salts	↓ hypertension, improved cognition
Time-restricted feeding [66, 119]	16–18 h fasting intervals	↑ autophagy, circadian alignment

Abbreviations: LDL-C, low-density lipoprotein cholesterol. Symbols: ↑ indicates increase or improvement; ↓ indicates decrease or reduction

Health Organization (WHO) recommendations, with even lower levels reported in Okinawa (≈1,113 mg/day) and a low prevalence of hypertension among Italian centenarians [130, 133–134]. Excess sodium is associated with cognitive decline and mortality, whereas potassium-enriched salts reduce cardiovascular risk [128, 129, 135, 136].

Moderate caloric intake and cultural practices such as *hara hachi bu* result in lower body mass index (BMI) and mild CR [127, 137]. Protein intake also follows a distinctive pattern. Protein intake averages around 18.5% of total energy [130], predominantly from plant sources, alongside high fiber consumption that supports metabolic and inflammatory health [138]. Fat intake is moderate (around 29%), mainly from mono- and polyunsaturated fatty acids [130, 138]. The New Nordic Diet (NND) reflects these principles in a contemporary regional model, emphasizing whole grains, root and leafy vegetables, wild berries, legumes, nuts, seaweeds, and cold-water fish, with rapeseed oil as the primary fat source. Evidence links it to improved cardiometabolic health and reduced chronic disease risk [67, 104].

Many centenarian populations also follow structured meal timing with prolonged nightly fasting (16–18 hours), supporting circadian and metabolic function [66]. Diets (Chinese) are further enriched in protective micronutrients, including copper, selenium, and manganese, contributing to metabolic resilience [139].

Collectively, these shared characteristics align with well-studied longevity-promoting patterns such as the Mediterranean [104, 140], Okinawan [67, 104], vegetarian/vegan [104, 141], and Nordic and the NND [67, 104], underscoring the central role of plant-centered, minimally processed diets in healthy aging and exceptional longevity.

Chronotype and chrononutrition

Chrononutrition examines how meal timing, frequency, and regularity interact with circadian biology to influence metabolic health [142, 143]. Because key metabolic tissues follow 24-hour rhythms, food intake acts as a zeitgeber capable of entraining these cycles. Late-day energy intake is linked to higher risks of obesity, impaired weight-loss response, cancer, and poorer glycemic control [142–151]. Cardiometabolic risk, including coronary heart disease and hypertension, has also been found to be elevated among individuals who regularly skip breakfast or consume late-night meals [152]. Emerging evidence further links misaligned eating patterns to accelerated biological aging [153]. These findings suggest that modifying eating patterns in alignment with circadian biology may represent a practical and cost-effective strategy to promote healthy aging and longevity.

Sleep quality

Sleep is a crucial determinant of health and longevity, with insufficient or poor-quality sleep associated with frailty [154], cognitive decline, and dementia [155, 156]. Sleep disturbances reduce muscle strength [157], in older adults [158], and impair metabolic health [159]. Increasing the risk of obesity, insulin resistance, type 2 diabetes, and cardiovascular disease [159–161]. Circadian disruption further promotes systemic inflammation and metabolic dysfunction [161, 162]. Optimizing sleep may therefore help preserve functional capacity, reduce chronic disease burden, and support healthy aging.

Maintaining an appropriate body weight

Centenarians are frequently described as lean, suggesting lower fat mass may contribute to longevity. Nevertheless, low body weight in older adults often reflects malnutrition [163], and standard BMI cut-offs may not apply to this age group [164]. The BMI-mortality association follows a U-shaped [165], with underweight status increasing mortality and functional impairment, while overweight may provide some protection against functional decline. Importantly, this protective effect appears to be most relevant in individuals over 80 years [166]. Unintentional weight loss greater than 5–10%, require close monitoring [167].

Ageing is accompanied by loss of muscle mass, increased adiposity, and ectopic fat deposition [168, 169] all of which elevate cardiometabolic risk [170]. Despite these changes, many centenarians present fewer chronic diseases and slower health decline before age 85 [171].

Accurate body composition (BC) assessment is critical. While gold-standard methods such as dual-energy X-ray absorptiometry and magnetic resonance imaging are costly and less feasible in routine care [172], bioelectrical impedance analysis (BIA) offers a portable, non-invasive,

Table 4 Nutritional and Lifestyle strategies for reaching 100 years

Strategy	Practical Implementation	Expected Benefit
Plant-based, nutrient-dense diet [67, 104]	Emphasize legumes, grains, vegetables, fruits	↓ chronic disease, ↑ lifespan
Sodium moderation [129]	< 2 g/day, potassium-enriched salt	↓ BP, ↓ CVD risk
Time-restricted feeding [124–129, 200, 201]	16:8 fasting or early dinners	↑ metabolic flexibility
Maintain lean mass [176]	BIA monitoring; adequate plant protein	↓ frailty
Daily physical activity [179, 202]	Walking, gardening	↑ cognition, ↓ inflammation
Adequate sleep [156, 159]	7–8 h nightly, consistent schedule	↓ insulin resistance, ↓ cognitive decline

Abbreviations: BP, blood pressure; CVD, cardiovascular disease; BIA, bioelectrical impedance analysis. Symbols: ↑ indicates increase or improvement; ↓ indicates decrease or reduction

cost-efficient alternative [173, 174], with potential value for evaluating nutrition and hydration in centenarians [173–176]. Sex-related differences in BC and hydration further underscore the need for individualized assessment [177].

Physical activity in centenarians

Physical activity remains a key component of healthy ageing and longevity, even among centenarians [178]. Daily low-intensity movement, walking, agricultural work, gardening, and navigating hilly terrain, is characteristic of long-lived populations (Nicoya Peninsula, Ikaria and Sardinia) and supports functional capacity, muscle preservation, and quality of life [179].

Regular activity also promotes cognitive health [179] by enhancing brain plasticity [180–183], mitochondrial function [184–186], redox balance [187–191], and neurotrophic signaling [192], thereby reducing cognitive decline. Exercise-induced gains in cardiorespiratory fitness support, brain volume [193, 194], and executive function in older adults [195]. The Rush Memory and Aging Project and other studies consistently show that regular physical activity enhances cognitive performance and resilience, including episodic memory [196–199].

Walking and regular exercise provide accessible, sustainable neuroprotection, reducing age-related disease risk and promoting healthy aging.

Evidence across centenarian studies, clinical trials, and population cohorts converges on actionable strategies for healthy aging: maintain a plant-rich, low-sodium, calorie-moderate diet; synchronize meals with circadian rhythms; preserve muscle through activity; and ensure restorative sleep. Integrating these factors forms

a multidimensional framework for achieving exceptional longevity (Table 4).

Nutraceuticals

To date, comprehensive reviews specifically addressing the use of nutraceuticals and dietary supplements among centenarians remain scarce. Most studies on lifestyle determinants of exceptional longevity highlight dietary rather than supplementation, and available data suggest historically low supplement use in these populations [130]. A meta-analysis in frail older adults similarly found no significant benefits of oral nutritional supplements on malnutrition or clinical outcomes [203].

Recent data from the Chinese Longitudinal Healthy Longevity Survey report low supplement use among centenarians (10.7% of women and 12.3% of men) [204], with calcium, protein, and multivitamins being the most commonly consumed; other nutrients were used by fewer than 2% of participants [204]. The limited adoption of supplements among centenarians suggests that the foundations of longevity may rely less on isolated nutraceutical interventions and more on long-term adherence to balanced diets and sustainable lifestyles. Nutraceuticals may serve as helpful adjuncts in cases of documented deficiency, but their broader role in healthy aging remains uncertain and requires more robust evidence.

Public health implications

Global demographic trends indicate rapid populations aging, with dramatic increases expected in individuals over 80 years of age [205, 206]. While longer lifespan reflects major public health progress, it also raises concerns about chronic diseases burden, disability, and health care costs [205, 206]. Evidence from centenarians shows that extended life expectancy does not necessarily lead to prolonged morbidity when diet and lifestyle are optimized [207], offering a model for compressing morbidity and prolonging functional years [205, 206]. This approach aligns with the WHO’s concept of “healthy longevity”, emphasizing autonomy, resilience, and quality of life in older age [205, 206].

Centenarian diets [130, 208], characterized by moderate caloric intake, high consumption of plant-based and minimally processed foods, and limited intake of animal fats and refined sugars, provide a strong foundation for preventive nutrition policies [40, 130, 208]. Their consistency across different cultural contexts underscores their universal applicability [130, 208]. These dietary patterns align with international recommendations for reducing cardiovascular diseases, diabetes, and cancer risk [209]. Integrating these insights into national dietary guidelines can reinforce the shift from disease treatment to healthspan promotion.

Portion control and mindful eating, common among centenarians, may help counter rising obesity rates in modern societies [66, 210]. Equally important are environmental determinants: centenarians often live in settings that facilitate access to fresh, locally grown foods, and limited exposure to ultra-processed products [138]. Public health interventions could emulate these environments through farmers' markets, urban agriculture, and subsidies for plant-based dietary staples [138].

Policy frameworks should also promote sustainability, as centenarians diets are environmentally friendly and align closely with planetary-health nutrition [67]. This alignment strengthens the case for incorporating nutrition into wider strategies that address climate change and food security [67].

Diet alone does not explain extreme longevity. Favorable socioeconomic conditions, strong community cohesion, supportive social structures [211], and equitable access to healthcare are recurrent features of regions with high concentrations of centenarians [212, 213]. Thus, dietary interventions must be paired with strategies addressing social determinants of health to maximize impact.

Translation of centenarian dietary patterns into modern public health practice requires cultural adaptation. While replicating traditional diets may not always be feasible, their core principles, simplicity, moderation, plant-based foundations, and social eating, are widely adaptable [130, 208]. Public health campaigns can focus on small, sustainable changes rather than rigid dietary models [130, 208, 214, 215].

Finally, investment in longitudinal research and centenarian registries is fundamental for refining precision nutrition approaches for older populations [216, 217].

Incorporating centenarian-inspired dietary principles into national health policies may help address the dual challenges of increasing life expectancy and rising non-communicable diseases ultimately promoting healthier aging across societies [130, 208].

Conclusion

The present review underscores the central role of nutrition in modulating biological pathways associated with longevity, particularly among centenarian populations. Evidence indicates that micronutrients such as vitamins, trace elements, and polyphenolic compounds play key roles in regulating oxidative stress, maintaining genomic integrity, and supporting mitochondrial function, ultimately helping to slow the development of age-related diseases. Dietary patterns consistently observed in longevity hotspots (including the Mediterranean, Okinawan, Nordic, and Nicoyan diets) are characterized by high consumption of plant-derived, nutrient-dense foods, moderate caloric intake, and limited reliance on animal

fats or processed products. These nutritional strategies converge on common mechanisms, notably reduced systemic inflammation, improved metabolic flexibility, and enhanced autophagic activity. Furthermore, CR and IF appear to potentiate nutrient-sensing pathways such as AMPK and sirtuin signaling, mitigating anabolic overstimulation via mTOR and supporting cellular repair and resilience. Notably, the gut microbiota emerges as an additional determinant of health span, with centenarians exhibiting a favorable microbial composition enriched in taxa such as *Akkermansia* and *Bifidobacterium*, which promote barrier integrity, immune modulation, and metabolic homeostasis. Nutritional interventions that sustain microbial diversity, particularly fiber- and polyphenol-rich foods, are therefore likely to contribute synergistically to healthy aging. From a broader perspective, these findings suggest that the convergence of molecular nutrition, microbial ecology, and cultural dietary practices underlies the remarkable health trajectories of centenarian populations. While genetic predisposition remains a non-modifiable determinant, the cumulative evidence strongly indicates that diet exerts a decisive influence on both lifespan and health span.

It becomes apparent that the study of centenarian diets offers a unique translational model for public health. However, it must be acknowledged the complexity of disentangling dietary effects from confounding lifestyle factors such as physical activity, psychosocial well-being, and environmental exposures, which often co-exist within these populations. Future research should move toward integrative, longitudinal designs incorporating nutrigenomics, metabolomics, and microbiome profiling to delineate causal pathways with greater precision. Moreover, public health recommendations should balance the promotion of nutrient-rich, plant-forward dietary patterns with pragmatic considerations of cultural acceptability, sustainability, and equity of access. Ultimately, the intersection of nutritional science and geroscience represents a fertile domain for interventions aimed not only at prolonging life, but also at ensuring its quality across aging populations.

Author contributions

E.F.-T. and C.R.-G.; conceptualization, E.F.-T., C.R.-G. and L.B.; methodology, E.F.-T. and C.R.-G.; data curation, L.V. and M.G.; formal analysis, C.R.-G.; investigation, G.A., E.J.-F., K.S. and M.G.; writing—original draft preparation, E.F.-T. and C.R.-G.; writing—review and editing, all authors; visualization, L.V. and M.G.; supervision, D.S.-R., G.M. and M.D.; project administration, C.R.-G. and D.S.-R.; validation, all authors; final approval, all authors have read and agreed to the published version of the manuscript and take responsibility for the accuracy and integrity of the content.

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Competing interests

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